

## **Part D**

### **Executive Summary**

The FreeInsuranceClaims100M dataset was used to evaluate two predictive modeling techniques: Random Forest (Model 2) and Logistic Regression (Model 1). Using the SS1 dataset from SmartShop Online, the analysis aims to find the model with the best predictive performance, spot areas for additional model optimization, and evaluate the changes needed for implementation.

Random Forest significantly outperformed Logistic Regression across the main evaluation metrics, according to the empirical results found in Part C. In comparison to the accuracy of 0.78 and ROC-AUC of 0.81 for Logistic Regression, Model 2 obtained an accuracy of 0.89 and a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) score of 0.94.

These results are in line with empirical studies showing how well ensemble machine-learning approaches work in challenging insurance classification issues (Brati et al., 2025). As a result, Random Forest is suggested as the best predictive model, subject to suitable validation, optimization, interpretability evaluation, and testing using real operational data from SmartShop Online.

### **1. Comparative Evaluation and Model Selection**

The outcomes in Part C show that Random Forest and Logistic Regression perform significantly differently in terms of prediction. A ROC-AUC of 0.81 and an accuracy of 0.78 were attained by logistic regression. The model was relatively less successful in capturing the more intricate relationships found in the dataset, despite the fact that these results show a reasonable discriminatory capability.

This restriction makes sense theoretically since Logistic Regression uses a predetermined functional structure to estimate relationships, which may necessitate the use of explicit feature transformations or interaction terms to represent intricate non-linear relationships. However, when model transparency and ease of interpretation are crucial factors, its relative simplicity is still beneficial.

By contrast, Random Forest obtained ROC-AUC of 0.94, accuracy of 0.89, precision of 0.82, and recall of 0.77. These findings show better overall classification performance as well as an enhanced capacity to recognize observations that belong to the positive class.

The results are largely in line with those of Brati et al. (2025), who compared a number of classification algorithms for the purpose of forecasting expensive insurance claims. According to their research,

Random Forest had the best classification accuracy (0.8867) and AUC (0.9437) of all the models tested, while Logistic Regression had the worst. Additional factors that the authors found to be significant predictors of high-cost claims included claim type, driver age, vehicle brand, and deferred period (Brati et al., 2025).

Random Forest (Model 2) is thus suggested for additional development and validation based on the empirical data gathered in Part C. Nevertheless, the suggestion should not be taken as proof that Random Forest will automatically generate comparable results when used with the SS1 dataset from SmartShop Online. The features, quality, quantity, and distribution of the data used to train and assess the model all affect its performance.

## **2. Opportunities for Model Improvement**

Even though Random Forest performs relatively well, there are a few methodological changes that could improve its generalizability and predictive reliability.

Hyperparameter optimization should be done first. Model performance can be affected by parameters like the number of trees (`n_estimators`), maximum tree depth (`max_depth`), minimum samples needed to split a node (`min_samples_split`), and number of features taken into account at each split. To find a better parameter configuration, methods like grid search, randomized search, or Bayesian optimization could be used in conjunction with cross-validation.

Second, the issue of class imbalance needs special attention. The class of greatest business interest may occur much less frequently than the majority class in fraud-detection and related classification applications. In these situations, an evaluation of model effectiveness based solely on accuracy may be insufficient. As a result, when assessing model performance, precision, recall, F1-score, ROC-AUC, and the confusion matrix should all be taken into account. When appropriate, resampling techniques like cost-sensitive learning or the Synthetic Minority Oversampling Technique (SMOTE) may also be looked into.

Third, by modifying current variables or adding new variables that more accurately capture the underlying phenomenon, feature engineering may enhance predictive performance. Interaction variables, customer behavioral metrics, claim ratios, transaction frequencies, and temporal variables, for instance, may show patterns that the original features did not fully capture.

Lastly, different ensemble techniques could be assessed as reference models. For instance, in their insurance claims study, Brati et al. (2025) discovered that XGBoost was the second-best classifier, behind

Random Forest. Therefore, testing other ensemble methods would offer more proof that the chosen model is a suitable solution for the particular dataset.

### **3. Adaptation of the Model to the SS1 Dataset**

Careful methodological adaptation is needed when moving from the FreeInsuranceClaims100M dataset to the SS1 dataset from SmartShop Online. It is not reasonable to assume that a model trained on one dataset will generalize well to another dataset with different underlying variables, populations, distributions, and business contexts.

Thus, data and schema alignment is the first prerequisite. The variables used in the development of the model should be mapped against the variables found in SS1. Numerical variables should go through the same pertinent preprocessing steps used during training, missing values should be handled methodically, and categorical variables should be encoded consistently.

Second, domain-specific feature engineering needs to be done. SS1 may offer useful predictive information if it includes variables like purchase frequency, transaction value, customer tenure, product returns, participation in loyalty programs, payment behavior, or past claims. However, rather than relying only on presumptions about business relevance, feature selection should be backed by exploratory analysis and validation.

Third, the Random Forest model's complexity needs to be adjusted for SS1's size and features. Unrestricted tree depth or excessive model complexity may raise the risk of overfitting if SS1 has significantly fewer observations than the original modeling dataset. As a result, cross-validation should be used to evaluate out-of-sample performance.

### **4. Interpretability and Alignment with the Literature**

The relationship between predictive performance and interpretability is an important factor in the adoption of machine-learning models. Compared to traditional statistical models, complex machine-learning models can produce highly predictive outcomes but offer less immediately clear explanations.

Using vehicle telematics data from 14,642 vehicles and roughly 125 million driver-trip observations, McDonnell et al. (2026) recently investigated this problem. In their study, explainable artificial intelligence (XAI) methods and various AI techniques were compared with conventional modeling approaches. In their insurance-claims context, the authors discovered that XAI techniques can help interpret complex

predictive models and found that XGBoost in conjunction with tools like Dalex or ExplainerDashboard is especially helpful (McDonnell et al., 2026).

This has practical implications for SmartShop Online. If Random Forest is adopted, interpretability techniques such as SHAP (SHapley Additive exPlanations) can be incorporated to explain how individual predictors contribute to model outputs. Such explanations may improve stakeholders' understanding of model behaviour and facilitate scrutiny of automated predictions.

The results of the present analysis are also consistent with Brati et al. (2025), whose empirical investigation demonstrated the potential of Random Forest for insurance claim classification. Collectively, the literature indicates that ensemble machine-learning approaches can provide substantial predictive capabilities, although model selection should also consider interpretability, data characteristics, operational requirements, and the consequences of classification errors.

## **5. Conclusion and Recommendations**

On the dataset examined in Part C, the comparative analysis shows that Random Forest (Model 2) outperforms Logistic Regression (Model 1) in terms of predictive performance. With an accuracy of 0.89, ROC-AUC of 0.94, precision of 0.82, and recall of 0.77, Random Forest proved to be more capable of classification in the current experimental environment.

Based on these results, Random Forest should be the main candidate model for future development at SmartShop Online. However, the model should go through feature engineering, cross-validation, class-imbalance analysis, systematic hyperparameter optimization, and independent testing using the SS1 dataset before being put into use.

The company should also implement suitable model-interpretability strategies and set up a monitoring system to identify shifts in the underlying data distributions and model performance. Because shifts in consumer behavior, transaction patterns, and other predictor distributions may eventually diminish model efficacy, ongoing monitoring is especially crucial.

All things considered, the data points to Random Forest as a promising predictive method for the current application. However, the quality and representativeness of SS1, suitable validation techniques, model interpretability, and continuous performance monitoring will all be necessary for successful implementation in addition to predictive accuracy.

## References

Brati, E., Braimllari, A., & Gjeçi, A. (2025). Machine learning applications for predicting high-cost claims using insurance data. *Data*, 10(6), 90. <https://doi.org/10.3390/data10060090>

McDonnell, K., Sheehan, B., & Murphy, F. (2026). Bridging transparency in insurance claims prediction: A comparative study of explainable AI and traditional linear models using vehicle telematics data. *Technological Forecasting and Social Change*, 223, 124418. <https://doi.org/10.1016/j.techfore.2025.124418>